PATENT APPLICATION

PROVISIONAL APPLICATION FOR LETTERS PATENT

TITLE: Self-Sustaining Neural-Motor Energy Harvesting Loop

for Autonomous Cognitive Systems

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GitHub: github.com/DarkWinD90/Consciousness_Env (tag v3.0.0-phase10-multimodal, commit 9e2c333)

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to co-pending provisional patent application for "Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation" and co-pending provisional patent application for "Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers," both by the same inventor, Kevin Christopher Ward. Each co-pending application is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to neuromorphic computing systems and, more particularly, to a self-sustaining neural-motor energy harvesting loop that integrates a spiking neural network with mechanical actuation and multi-modal energy harvesting in a closed-loop architecture that exhibits homeostatic energy regulation and recursive self-observation.

BACKGROUND OF THE INVENTION

[0003] Neuromorphic computing systems that implement spiking neural network (SNN) dynamics require sustained power to maintain membrane potentials, process spike trains, and update synaptic weights through spike-timing dependent plasticity (STDP). Conventional neuromorphic systems draw continuous power from fixed external supplies, creating a dependency on infrastructure that limits deployment in autonomous, embodied, or resource-constrained contexts.

[0004] Energy harvesting techniques have been applied to extend the operational lifetime of wireless sensor nodes, implantable medical devices, and small-scale robotics. Piezoelectric transducers convert mechanical strain and vibration into electrical energy. Thermoelectric generators (TEGs) convert thermal gradients between hot and cold surfaces into electrical current. Triboelectric nanogenerators (TENGs) harvest energy from relative motion between surfaces of dissimilar materials.

[0005] However, prior systems treat energy harvesting as a passive supplement to primary power, not as a mechanism for active homeostatic regulation. No prior system creates a closed-loop architecture in which the neural processing activity itself generates the mechanical motion that drives energy recovery back into the neural processing substrate. The invention described herein addresses this gap.

[0006] Prior work in physics-informed neural networks, such as EPFL's Dynami-CAL GraphNet described in Nature Communications (February 2026), embeds Newton's third law into Graph Neural Networks for rigid-body physics simulation. That work is fundamentally different in mechanism (constraint embedding versus self-referential feedback), substrate (Graph Neural Networks versus Spiking Neural Networks), and purpose (stable physics simulation versus embodied consciousness). The present invention's hardware claims (directed to energy harvesting from servo movement via triboelectric and pyroelectric transduction) and its recursive reflection claims (directed to self-observation in neuromorphic systems) have no overlap with that prior art.

SUMMARY OF THE INVENTION

[0007] The present invention provides a self-sustaining neural-motor energy harvesting loop for autonomous cognitive systems. The loop comprises a spiking neural network that processes sensory inputs and generates motor commands, a motor that converts neural spike patterns into mechanical actuation, an energy harvesting subsystem that recovers energy from the mechanical motion through piezoelectric and thermoelectric transduction, and a recursive self-observation layer that feeds prior output back as input on the next processing step.

[0008] The key insight is that the energy harvesting yield depends on servo movement variance, not absolute position. A network of N leaky integrate-and-fire neurons with n active spikes per step generates output variance proportional to the binomial variance $n(N-n)/N$, which peaks at $n = N/2$. This creates a bell-shaped (sinusoidal) relationship between neural activity and energy harvest, establishing a bounded surplus zone between a lower and upper crossover threshold. The system exhibits homeostatic self-regulation by maintaining neural activity within this surplus zone, constituting a form of metabolic self-regulation that is an emergent property of the coupled architecture.

[0009] A recursive reflection layer compares the system's predicted state to its observed state and drives adaptive behavior through a depth-limited self-observation feedback loop with a configurable reflection coefficient. The metabolic cost of consciousness is modeled: too little activity drains stored energy due to base consumption exceeding harvest; too much activity drains energy due to the quadratic cost penalty exceeding the declining harvest at high spike rates. A Goldilocks zone exists in which neural activity sustains itself through homeostatic energy regulation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate preferred embodiments of the invention.

[0011] FIG. 1 is a system architecture block diagram showing the self-sustaining neural-motor energy harvesting loop with a sensor 12, spiking neural network 14, motor 16, power management block 18, thermal harvester 20, piezoelectric element 22, ground reference 24, control logic block 26, energy store 28, and reflection feedback path 30 interconnected in a closed loop within an environment boundary 10.

[0012] FIG. 2 is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2,000 operational steps, plotted on a y-axis 32 and x-axis 34 with an origin point 38 at 50 mWh, a zero energy baseline 40, an experimental curve 36 with experimental curve label 42 and experimental legend box

46, and a control curve 44 with control curve label 50 and control legend box 48.

[0013] FIG. 3 is a schematic diagram of the spiking neural network 14 architecture showing the input layer 52, weight matrix 54, hidden layer 56 with leaky integrate-and-fire neurons and neuron detail section 58, output stage 60, spike output 62, membrane potential dynamics 64, spike vector output 66, and STDP learning rule box 68.

[0014] FIG. 4 is a circuit schematic of the energy harvesting subsystem within a circuit boundary 70 comprising a piezoelectric element 22 path with AC signal 72, inductor 74, rectifier 76, and smoothing capacitor 78, and a thermoelectric path with heat sink 80, TEG module 82, and boost converter 84, both feeding through an output regulator 86 and system output 88 to the spiking neural network 14.

[0015] FIG. 5 is a graph showing activity-dependent energy dynamics within a main graph area 90, including a sinusoidal harvest curve 92, quadratic cost curve 94, metabolic operating point 96, deficit zones 98, optimal operating range 100, surplus zone 102, optimal zone boundaries 104, governor feedback loop 106, energy-aware modulation sub-graph 132, operating point equations 134, lower crossover point 136, and upper crossover point 138.

[0016] FIG. 6 is a hardware reference design layout diagram showing the microcontroller 108, piezoelectric element 22, motor 16, thermistor 110, photoresistor 114, LED indicator 112, AC signal 72, control logic block 26, and energy store 28 with their interconnections.

[0017] FIG. 7 is a validation results summary comprising a claims validation table 116, an energy comparison table 118, an energy trajectory graph 120, a legend 122, and a summary box 124.

[0018] FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the signal flow from an environment input arrow 126 through sensor interface 12, cognitive modulation block 128, spiking neural network 14, motor 16, thermal harvester 20, analog-to-digital converter (ADC) 130, piezoelectric element 22, and energy store 28, with reflection feedback 30, all within the environment boundary 10. The "ADC" abbreviation is used interchangeably with "analog-to-digital converter" throughout this specification and the drawings.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

I. System Architecture Overview

[0019] Referring to FIG. 1, the self-sustaining neural-motor energy harvesting loop operates within an environment boundary 10. The environment boundary 10 is a logical enclosure representing the complete system, including all energy transduction, neural processing, and motor actuation elements. The system implements a closed-loop signal and energy flow: SENSE → PROCESS → ACTUATE → HARVEST → REGULATE → REFLECT.

[0020] A sensor 12 receives environmental input, which may include light intensity from photodiode arrays, temperature from thermistor elements, vibration from accelerometers, or other physical stimuli. The sensor 12 converts the physical input to an electrical signal suitable for input to the spiking neural network 14.

[0021] The spiking neural network (SNN) 14 processes the sensor input through a population of leaky integrate-and-fire (LIF) neurons. In the preferred embodiment, the SNN 14 comprises N=50 neurons organized in an all-to-all connected topology with synaptic weights bounded to [0.0, 0.5]. Membrane potential dynamics follow $V(t+1) = V(t) \times \text{leak} + I$, where leak = 0.95 provides sub-threshold decay between spikes. A neuron fires when $V \geq V_{th} = 0.5$, after which V resets to zero. Synaptic weights update through spike-timing dependent plasticity (STDP) per the rules detailed in FIG. 3 and paragraph

0036.

[0022] A motor 16 converts the aggregate SNN spike output into mechanical actuation. In the preferred embodiment, the motor 16 is a servo motor (SG90 or equivalent) controlled by pulse-width modulation from the microcontroller 108. The motor 16 drives a rotating shaft that maintains mechanical coupling to the piezoelectric element 22.

[0023] A thermal harvester 20 captures thermal energy generated by system operation. Thermal differentials between the SNN processing substrate and the ambient environment drive a thermoelectric generator (TEG module 82) that contributes to the energy store 28 through the boost converter 84 and output regulator 86.

[0024] A piezoelectric element 22 (27mm piezo disc in the preferred embodiment) converts mechanical vibration from the motor 16 into electrical energy via the piezoelectric effect. The motor shaft couples to the piezo element 22 through a nearly-locked joint configuration that maximizes strain energy per actuation cycle. An LC resonance circuit (inductor 74) amplifies the AC voltage output (AC signal 72) by 3-10× at the resonant frequency before rectification.

[0025] A control logic block 26 manages the distribution and regulation of harvested energy. The control logic block 26 is implemented as decision logic in the microcontroller 108 firmware, evaluating the current energy store level and the computed modulation coefficient to determine whether to permit, throttle, or amplify SNN activity.

[0026] An energy store 28 buffers harvested energy for sustained operation. In the preferred embodiment, the energy store 28 is a supercapacitor bank with a target storage capacity of 100 mWh. The energy store 28 provides power to the SNN 14 through the power management block 18 and maintains sufficient reserve for burst activity when the system encounters high-priority stimuli.

[0027] A reflection feedback path 30 implements recursive self-observation by feeding the prior neural output $r(t) = y(t-1)$ back as an additional input to the SNN 14 on the next processing step. The reflection feedback path 30 has a configurable reflection coefficient ($\rho \in [0, 1]$) that scales the contribution of the prior output to the current input. When $\rho = 0$, the system operates without self-observation. When $\rho > 0$, the SNN 14 processes a weighted mixture of external sensor input and its own prior output, creating recursive self-referential dynamics.

II. Energy Model — Activity-Dependent Dynamics

[0028] The energy balance for each simulation step is computed as: $\Delta E = E\_harvest(n) - E\_consume(n)$, where n represents the number of neural spikes per step. The harvest term E_harvest(n) is a sinusoidal function of neural activity: $P\_harvest(n) = H\_max \times \sin(\pi n / N)$ milliwatts, where H_max = 121 mW and N = 50 (number of SNN neurons). This functional form arises because energy harvesting depends on servo movement variance rather than absolute position; movement variance follows binomial statistics with a peak at $n = N/2$, and the sine function provides the continuous approximation. The consumption term E_consume(n) increases quadratically with activity: $P\_consume(n) = 45 + 1.2n + 0.06n^2$ milliwatts, where 45 mW is base system consumption, 1.2 mW is per-spike activity cost, and 0.06 mW is the quadratic burst penalty reflecting superlinear scaling of neural computation cost with firing rate.

[0029] The energy model parameters are implemented in the BalancedEnergyConfig configuration class: base_consumption_mw = 45.0; activity_cost_mw = 1.2; activity_cost_quadratic = 0.06; time_step_hours = 0.005; storage_capacity_mwh = 100.0; initial_energy_mwh = 50.0; self_discharge_rate = 0.001. These parameters produce the activity-energy dynamics depicted in FIG. 5, where the harvest curve follows a sinusoidal profile peaking at 121 mW due to servo movement variance, and the cost curve follows the quadratic consumption formula.

[0030] Referring to FIG. 5, the interaction between the sinusoidal harvest curve 92 and the quadratic cost curve 94 creates a characteristic energy dynamic with a bounded surplus region 102 and a metabolic operating point 96. At rest state of 0 spikes, there is a deficit 98 of approximately 45 milliwatts due to base system consumption with no harvesting. At low activity below approximately 8 spikes, cost exceeds harvest as servo movement variance is insufficient to overcome base consumption. At medium activity from approximately 8 to 27 spikes, the surplus zone 102 exists where harvest exceeds cost; the peak surplus of approximately 23 milliwatts occurs near 18 spikes. The metabolic operating point 96 occurs at approximately 14 spikes where harvest of approximately 94 milliwatts equals cost of approximately 74 milliwatts plus self-discharge losses of approximately 20 milliwatts. The optimal operating range 100 spans approximately 10 to 25 spikes, bounded by optimal zone boundaries 104. Above approximately 27 spikes, the quadratic cost penalty dominates and the system enters the deficit zone 98 where cost exceeds harvest. The energy-aware modulation sub-graph 132 shows how the governor feedback loop 106 adjusts neural activity to maintain the system within the surplus zone, constituting a form of metabolic self-regulation as described by the operating point equations 134.

III. Spiking Neural Network Architecture

[0031] Referring to FIG. 3, the spiking neural network 14 comprises an input layer 52, a weight matrix 54, a hidden layer 56 with N leaky integrate-and-fire neurons and accompanying neuron detail section 58, an output stage 60 that generates spike output 62, a membrane potential dynamics display 64, a spike vector output 66, and an STDP learning rule box 68.

[0032] The input layer 52 accepts two signal types: external sensory input from the sensor 12, and the recursive self-observation signal from the reflection feedback path 30. This dual-input architecture enables the SNN 14 to process environmental stimuli while simultaneously monitoring its own prior output, creating the conditions for emergent self-referential behavior.

[0033] The weight matrix 54 contains all-to-all synaptic connections W[i,j] with weights bounded to [0.0, 0.5]. The upper bound of 0.5 prevents runaway excitation and enforces stability. Synaptic weights update continuously through the STDP learning rule 68: potentiation (LTP) occurs when a post-synaptic neuron fires shortly after a pre-synaptic neuron (W += A+ × pre_trace); depression (LTD) occurs when a pre-synaptic neuron fires shortly after a post-synaptic neuron (W −= A− × post_trace). The stability bias A− > A+ ensures the network does not develop runaway synchrony.

IV. Energy Harvesting Circuit

[0034] Referring to FIG. 4, the energy harvesting subsystem is enclosed within a circuit boundary 70 and comprises two parallel energy recovery paths: a piezoelectric harvesting path and a thermoelectric harvesting path.

[0035] The piezoelectric harvesting path begins at piezoelectric element 22, which generates an AC signal 72 proportional to mechanical strain. An inductor 74 forms an LC resonance circuit with the piezoelectric element's intrinsic capacitance, providing voltage amplification at the resonant frequency. A full-bridge rectifier 76 converts the AC signal to DC. A smoothing capacitor 78 filters the rectified output before it enters the output regulator 86. The thermoelectric path recovers heat through heat sink 80, TEG module 82, and boost converter 84, converging at the output regulator 86 alongside the piezoelectric path. The regulated combined output delivers power to the SNN 14 through system output

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V. Validation Results

[0036] The system was validated computationally through a Python simulation across 2,000,000 operational steps. Referring to FIG. 7, five falsifiable claims were tested: (A) closed-loop continuity over 2,000 steps — PASS; (B) boundedness of all state variables — PASS; (C) noise robustness with standard deviation ≤ 45 — PASS; (D) input-output gain with correlation ≥ 0.2 — PASS; (E) saturation ratio ≤ 0.20 — PASS. Additionally, the claims validation table 116 records validation of nine further claims across three additional phases: Phase 8 (Synaptic Plasticity) tested weight entropy decrease (F8.1 — PASS), STDP mutual information ratio exceeding 1.20× (F8.2 — PASS), and convergence below 10% (F8.3 — PASS); Phase 9 (Predictive Processing) tested prediction error reduction exceeding 50% (F9.1 — PASS), absence of spurious trend in random baseline with $p > 0.05$ (F9.2 — PASS), and adaptation spike exceeding 1.3× with recovery (F9.3 — PASS); Phase 10 (Multi-Modal Integration) tested synchronization difference exceeding 0.20 (F10.1 — PASS), weight entropy decrease (F10.2 — PASS), and convergence below 0.20 (F10.3 — PASS). The summary box 124 confirms that all 14 of 14 claims passed across Phases 7 through 10. The energy comparison table 118 records the terminal energy states: BalancedEnergyConfig (experimental) at +100.0 mWh, EnergyConfig (control baseline) at −4,697 mWh. FIG. 2 shows the diverging energy trajectories of these two configurations over 2,000 steps: experimental curve 36 rising to storage capacity, control curve 44 depleting to negative values.

VI. Hardware Reference Design

[0037] Referring to FIG. 6, the preferred hardware embodiment uses a Raspberry Pi Pico (RP2040) microcontroller 108 as the central processing hub. The piezoelectric element 22 connects to the microcontroller 108 via the GP26 analog input pin (ADC0). The motor 16 connects via PWM output with a mechanical coupling to the piezoelectric element 22 for vibration harvesting. A thermistor 110 monitors temperature via GP27 (ADC1). A photoresistor 114 (GL5528 LDR) provides light sensing via GP28 (ADC2). An LED indicator 112 provides status visualization via GPIO. The control logic block 26 (decision diamond) governs energy distribution decisions. The energy store 28 (supercapacitor bank) connects to VSYS for system power delivery.

VII. Reference Architecture

[0038] Referring to FIG. 8, the energy-bounded recursive control architecture shows the complete signal flow within the environment boundary 10. Environmental signals enter via environment input arrow 126 to the sensor interface 12. The sensor interface 12 feeds analog-to-digital converter 130, which digitizes the sensor output for input to the spiking neural network 14. An optional cognitive modulation block 128 (shown as a cloud shape, representing external cognitive control) provides attention weighting and intention bias to the SNN core 14. The SNN core 14 processes the modulated input and drives motor output 16. The motor 16 drives mechanical actuation that feeds the thermal harvester 20 and piezoelectric element 22, which harvest energy into the energy store 28. Reflection feedback path 30 (containing the equation $r(t) = y(t-1)$) returns prior output as input to the SNN 14. The energy sustaining loop (bold dashed path) recycles harvested energy from the energy store 28 back to the SNN core 14, completing the metabolic loop.

CLAIMS

What is claimed is:

1. A self-sustaining neural-motor energy harvesting system comprising: a spiking neural network (SNN) configured to process sensory input through a population of leaky integrate-and-fire neurons and generate a spike output; a motor mechanically coupled to receive the spike output and convert neural spike patterns into mechanical actuation; an energy harvesting subsystem configured to recover electrical energy from the mechanical actuation through at least one of piezoelectric transduction and thermoelectric transduction; an energy store configured to buffer the recovered electrical energy and supply power to the SNN; and a control logic block configured to regulate the distribution of harvested energy based on current energy store level; wherein the SNN, motor, energy harvesting subsystem, and energy store form a closed energy loop in which the SNN's neural activity drives the mechanical actuation that generates the energy that sustains the SNN's neural activity.

2. The system of claim 1, wherein the energy harvesting subsystem comprises a piezoelectric element mechanically coupled to the motor output, an LC resonance circuit connected to the piezoelectric element for voltage amplification, a full-bridge rectifier connected to the LC resonance circuit, a smoothing capacitor connected to the rectifier output, and an output regulator connected to the smoothing capacitor and energy store.

3. The system of claim 1, wherein the energy harvesting subsystem further comprises a thermoelectric generator module configured to harvest thermal energy from a temperature differential between the SNN processing substrate and the ambient environment, and a boost converter connected to the thermoelectric generator module and the output regulator.

4. The system of claim 1, further comprising a recursive reflection layer configured to feed a prior spike output of the SNN back as an input to the SNN on a subsequent processing step, wherein the recursive reflection layer implements self-referential feedback with a configurable reflection coefficient $\rho \in [0, 1]$; and wherein the recursive reflection layer is implemented within the energy-harvesting closed loop such that the reflection computation draws power from the energy store.

5. The system of claim 4, wherein the recursive reflection layer is configured with a maximum recursion depth constraint to prevent unbounded self-referential depth; and wherein the reflection coefficient is dynamically modulated based on the current energy level of the energy store, reducing self-observation activity when the energy store approaches a critically low threshold.

6. The system of claim 1, wherein the control logic block implements an energy governor configured to: monitor a current energy storage level; compute a modulation coefficient from an energy-aware modulation function having at least four zones: a critical zone with suppression coefficient, a low zone with mild suppression, a normal zone with neutral coefficient, and a high zone with amplification coefficient; and apply the modulation coefficient to the SNN's input gain or firing threshold; wherein the SNN's neural activity self-regulates toward an activity level within a surplus zone bounded by a lower crossover point and an upper crossover point of the harvest-versus-cost power curves.

7. The system of claim 1, wherein the energy harvesting yield follows a sinusoidal function of neural activity $P\_harvest(n) = H\_max \times \sin(\pi n/N)$, where n is the spike count per step and N is the total number of SNN neurons, arising from the physical relationship between neural activity and servo movement variance, which follows binomial statistics peaking at $n = N/2$.

8. The system of claim 7, wherein the energy consumption follows a quadratic function P_consume(n) = b + c■n + c■n², where b is base consumption, c■ is per-spike linear cost, and c■ is a quadratic burst penalty; and wherein the intersection of P_harvest(n) and P_consume(n) + P_self_discharge defines a metabolic operating point at a spike count n* at which the system achieves homeostatic energy equilibrium.

9. The system of claim 7, wherein the sinusoidal harvest profile and quadratic cost profile together create an optimal operating point at medium activity levels where energy harvesting exceeds consumption, with both low-activity and high-activity states resulting in net energy drain, such that the system exhibits metabolic homeostasis analogous to biological neural metabolism.

10. A method for self-sustaining neural-motor energy harvesting comprising: processing sensory input through a spiking neural network to generate a spike output; converting the spike output to mechanical actuation via a motor; harvesting electrical energy from the mechanical actuation through piezoelectric and/or thermoelectric transduction; buffering the harvested electrical energy in an energy store; regulating neural activity using a governor feedback loop that modulates SNN firing based on current energy store level; and feeding a prior SNN output back as input on a subsequent processing step to implement recursive self-observation; wherein the harvested energy is sufficient to sustain the SNN operation across the closed loop.

11. A non-transitory computer-readable medium storing instructions that, when executed by a neuromorphic processor, cause the processor to: implement a leaky integrate-and-fire neuron population with N neurons, wherein membrane potential evolves as V(t+1) = V(t) × leak + I and a spike is generated when V ≥ V_th; compute an energy balance each processing step as ΔE = E_harvest(n) − E_consume(n), wherein E_harvest(n) = H_max × sin(πn/N) and E_consume(n) = b + c■n + c■n^2; apply a modulation coefficient to the SNN input gain based on current energy store level; and feed the prior spike output back as input with a reflection coefficient ρ; wherein the instructions implement autonomous energy homeostasis without external power regulation.

ABSTRACT

A self-sustaining neural-motor energy harvesting loop for autonomous cognitive systems integrates a spiking neural network (SNN) with motor actuation, piezoelectric and thermoelectric energy transduction, and a recursive self-observation layer in a closed-loop architecture. The system exploits the relationship between neural activity and servo movement variance, which follows binomial statistics and is approximated by a sinusoidal harvest function P_harvest(n) = H_max × sin(πn/N). This sinusoidal harvest profile, combined with a quadratic energy consumption cost P_consume(n) = 45 + 1.2n + 0.06n^2, creates a bounded surplus zone of neural activity within which the system achieves homeostatic energy equilibrium. A governor feedback loop monitors the energy store and dynamically modulates the SNN firing threshold to maintain activity within this Goldilocks zone. A reflection feedback path implements recursive self-observation by feeding prior output as input on each subsequent step, with configurable reflection coefficient and depth-limited recursion. The system was validated over 2,000,000 simulation steps achieving autonomous energy self-sufficiency. A hardware prototype using an RP2040 microcontroller, SG90 servo, and 27mm piezoelectric element implements the architecture at an approximate bill-of-materials cost of $475.

SPECIFICATION CHANGE LOG — v3 FIG. 5 UPDATE

¶0015 (Brief Description of FIG. 5) — REVISED:

Added numerals 132 (energy-aware modulation sub-graph), 134 (operating point equations), 136 (lower crossover point), and 138 (upper crossover point) to the brief description. These components existed in the figure but were not previously listed in the brief description.

¶0028 (Energy Balance Equation) — REVISED:

The prior ¶0028 referenced a linear harvest relationship. Revised to explicitly describe the sinusoidal form $P\_harvest(n) = H\_max \times \sin(\pi n/N)$ and its physical basis: energy harvest depends on servo movement variance, which follows binomial statistics with a peak at $n = N/2$. The quadratic consumption formula is unchanged (derived directly from ¶0029 parameters).

¶0030 (Zone Descriptions) — REVISED:

The prior ¶0030 listed stale zone boundaries (low: 5-15 spikes, medium: 30-50 spikes, high: 70-85 spikes) that were inconsistent with the validated model. Revised to the mathematically correct boundaries derived from the ¶0029 parameters: low-activity deficit (0 to ~8 spikes), surplus zone (~8 to ~27 spikes, peak +23 mW at n=18), high-activity deficit (~27 to 50 spikes). All 13 unified FIG. 5 reference numerals (90–138) are now cited in the paragraph prose. The operating point is defined as harvest = cost + self-discharge (~94 mW = ~74 mW + 20 mW at $n^* \approx 14$ spikes).

Claim 9 — VERIFIED, NO REVISION:

Claim 9 uses the language 'medium activity levels' without specifying spike counts. The corrected zone boundaries (8–27 spikes) fully satisfy the structural requirement of low-drain / medium-surplus / high-drain, so no claim language revision is required. The shift in zone boundaries does not affect claim scope.

All formulas independently verified: